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Consumer Online Shopping Behavior Prediction Based on Machine Learning Algorithm

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Abstract

Based on the data of real users, goods and consumption behaviors of e-commerce platforms, this paper uses data mining technology and machine learning algorithm to build a prediction model of consumers' online shopping behavior, so as to better analyze consumers' online shopping behavior, and provide data support for the realization of precision marketing of e-commerce enterprises. Through model verification and final model evaluation, the accuracy of the model is 0.91, indicating that the model can scientifically and accurately predict consumers' purchase behavior intention on the e-commerce platform in most cases.

Keywords: : Machine learning, Online shopping behavior, Prediction model, XGBoost

1. Introduction

In the context of the integration and development of digital economy and social media, Chinese social media platforms have realized the transformation to artificial intelligence technology paradigm. The continuous expansion of the user scale makes the recommendation algorithm system become the core infrastructure of the platform economy, and the construction of accurate prediction model and intelligent recommendation mechanism has become the key elements for enterprises to obtain competitive advantages. In the big data environment, the demand for multidimensional data processing capabilities shows an exponential growth trend. The information value discovery technology based on deep data mining, especially the intelligent analysis ability of user behavior patterns, is becoming an important research field of modern information science. In the initial development stage of e-commerce, enterprise business intelligence system mainly relies on customer transaction records to build. Its typical model is to collect periodic sales data (usually quarterly), combine revenue indicators and profit analysis to form

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business reports, and formulate phased marketing strategies to maintain the stability of business operations. However, this analysis paradigm based on explicit consumption data has significant limitations. On the one hand, the traditional consumption data analysis model is difficult to effectively capture the potential consumption intention and behavioral preferences of users; On the other hand, due to the lack of in-depth analysis of consumer consumption behavior, there is a precision bottleneck for enterprises to identify high-value customers, and there is a serious imbalance between the cost and output benefits of traditional manual data analysis. The emergence and breakthrough of machine learning algorithm technology has provided strong technical support for many industries, especially for the paradigm innovation of e-commerce industry decision-making system, providing powerful technical algorithms, such as the user cluster model based on Lasso regression and the consumer behavior prediction algorithm, through the establishment of a dynamic classification mechanism of multidimensional feature space. The visual subdivision of user groups and the simulation of consumption trajectory are realized. This technological evolution not only significantly improves the identification efficiency of high-value users, but also further improves the market response speed and consumer matching speed of e-commerce enterprises' marketing strategies through the dynamic optimization mechanism of consumer online shopping behavior prediction model, and finally builds a data-driven precision marketing closed-loop system.

2. Literature review

There are always two problems in the prediction of consumers' online shopping behavior, the first is the processing of unbalanced data, and the second is the selection of characteristics of consumers' online shopping behavior. For example, Wang Dong [1](2024) uses K-means cluster analysis to allocate the data of eliminated samples to each group according to the proportion of sample data in each group, thus achieving the research sample equilibrium. This method can effectively avoid the information loss caused by the random insufficient sample collection method. Liu Yifeng et al. [2](2024) proposed a combined network model of sample +CNN.LSTM under segmented conditions to test and analyze the real online shopping data of consumers on specific e-commerce platforms. The research results showed that the F1 value of this model increased by 7% and 11% on average compared with the standard analysis model. In order to more accurately discover the evolution of consumers' online shopping preferences, Luo Mi et al.[3] (2021) proposed a new, more secure federated collaborative filtering algorithm that is suitable for large-scale networks and can effectively integrate multiple information.

Liu Shuchang et al.[4] (2024) proposed a multi-source heterogeneous data integration framework, including API interface, log capture and web crawler technologies, and pointed out that data cleaning (such as outlier removal and missing value filling) and feature engineering (such as user activity index construction) are key steps to improve model performance. Taking an e-commerce platform as an example, this study reduced the original data dimension by 40% through feature dimensionality reduction technology (such as principal component analysis), and improved the model training efficiency by 2.3 times.

Peng Xinglin et al. [5](2024) further quantified the scale and value of user behavior data in "Research on Online Shopping User Behavior Based on Big Data Analysis", pointing out that the daily user behavior data of a certain e-commerce platform is as high as hundreds of millions of pieces, covering multidimensional information such as browsing, search and payment. The research emphasizes that association rule mining and time series analysis can reveal cross-category purchasing rules and periodic behavior patterns of users, and provide support for inventory management and promotion strategies.

Zhang Yuexian et al.[6] (2023) focused on the scenario of repurchase prediction and proposed an integrated learning framework. By integrating features such as user historical consumption frequency, customer unit price and return rate, the F1 score of repurchase prediction was increased to 0.87, 19% higher than that of a single logistic regression model. In addition, the study points out that model interpretive tools, such as SHAP value analysis, can help enterprises identify the core characteristics of high-value users.

Zhao Miao et al.[7] (2023) explored the balance between privacy protection and prediction accuracy. Developed FedBuy, a distributed prediction framework based on federated learning that allows users to store behavioral data locally, uploading only model gradient parameters. The framework achieves 92% of the performance of the centralized model on the recall index while maintaining data security, opening up a new path for compliance commercial applications.

Pan Qiuyu et al.[8] (2022) proposed MetaShop, a meta-learning framework, to realize adaptive prediction of cross-cultural consumer behavior. By constructing a hyperscale dataset containing six of the world's largest e-commerce platforms, their proposed prediction model has a zero-sample prediction accuracy of 81.4% in emerging markets in Southeast Asia, verifying the cultural adaptability potential of machine learning models.

Based on the existing research content and research direction, it can be seen that machine learning algorithm significantly improves the accuracy and practicability of consumer behavior prediction through feature engineering, model optimization and multi-source data integration. In the future, with the maturity of deep learning and edge computing technology, user behavior prediction will develop in the direction of real-time, intelligence and explainability, providing more powerful technical support for the fine operation of the e-commerce industry.

3. Research methods

3.1. Determination process of consumer purchasing behavior prediction model

This paper uses machine learning algorithm to build a prediction model for the prediction of consumers' online shopping behavior. Taking the sales data of an e-commerce platform as an example, the prediction model of consumers' purchase is built based on the real consumer users, online commodities, online shopping behavior data and other information of the platform. It uses data analysis and machine learning algorithms to predict consumers' future purchase intentions for a certain category of goods.

The typical feature of online consumption behavior data is the significant non-equilibrium of user behavior distribution. In the interaction process of the e-commerce platform, the click browsing record and shopping cart addition data generated by users present a massive scale, but the proportion of order data that actually triggers transactions is extremely limited. This typical category imbalance poses special challenges to data modeling: From the perspective of business value, the purchasing behavior user group with high conversion potential has core analytical value, and accurately identifying such target customer groups will directly improve the allocation efficiency of marketing resources[9]. It is worth noting that the asymmetry of this data structure is prone to predictive bias in machine learning models. When the dominant category (non-purchasing behavior) accounts for too high a proportion in the training set, the conventional algorithm naturally tends to preferentially fit most of the class samples in the process of maximizing the overall accuracy. In this case, even if there are significant errors in the identification of key minority classes (purchase behavior), the overall performance index of the model may remain high, resulting in misleading optimization results.

Therefore, to construct an effective prediction model, we must break through the limitations of traditional algorithms, and focus on solving the feature learning problem in the unbalanced category scenario. The ideal model needs to improve the classification sensitivity of the key minority classes while ensuring the accuracy of the majority classes, which poses dual challenges for algorithm optimization and evaluation index design[10]. In particular, it is necessary to establish a differentiated learning mechanism, so that the model can fully capture the transformation law between different behavior patterns, and finally achieve accurate hierarchical prediction of user purchase intention.

3.2. Prediction model construction and algorithm process

This paper adopts a machine learning algorithm based on decision tree integration, namely XGBoost, which is mainly applied to classification and regression problem analysis. Compared with the traditional GBDT algorithm, this learning algorithm has faster running speed, higher model generalization ability and higher prediction accuracy, and the algorithm can support multiple loss functions. For example, the square error and logarithm loss can meet the application requirements of multiple application scenarios.

First, we assume that there is a regression problem, which needs to solve a continuous target variable y. For the prediction task of continuous target variable, the gradient lifting method based on ensemble learning framework can build a regression tree model by optimizing strategies in stages. The algorithm builds an addition model iteratively, and its core mechanism is to guide the generation of a new base learner through the predicted residual of the current model. In the process of implementation, each iteration step follows the optimization path of the negative gradient direction of the loss function, and the asymptotic approximation to the optimal solution of the objective function is realized by the stepwise superposition weak learner.

The key implementation points are: first, the pseudo-residual is calculated based on the prediction error of the initial model, and then the residual term is used as the fitting target of the new regression tree. Through the double operation of error calculation and base model updating, the regression tree generated at each stage is weighted and combined to form an integrated prediction model with strong generalization ability. This progressive modeling strategy can effectively overcome the overfitting defect of single decision tree model, while retaining the advantages of tree model in feature interaction capture.

Initialize the model as: $f_0(x)$, for example, you can set it to the average of y, for t=1, 2, ..., T can be calculated according to the following steps:

1) Calculate the negative gradient:

$$gtx_i - = -\partial Ly_i, f_t - 1x_i \partial f_t - 1x_i \tag{1}$$

2) Fit a regression tree model $h_t(x)$ into the following formula:

$$\sum i = 1nLy_i, f_t - 1x_i + h_t x_i \tag{2}$$

3) The model is updated as follows:

$$f_t(x) = f_t - 1(x) + \eta h_t(x) \tag{3}$$

The core elements of gradient lifting algorithm include two key components: the selection of loss function and the regulation mechanism of learning rate. The former quantifies the prediction deviation through mathematical forms such as square error and absolute loss, while the latter uses shrinkage coefficient to adjust the contribution of a single regression tree to the integrated model, so as to balance the model complexity and generalization ability. The algorithm approximates the optimal solution space by correcting the residual error of the preorder model one round at a time. As an advanced implementation of the framework, XGBoost is systematically enhanced through three dimensions: First, the second derivative information of the loss function is introduced to build an optimal tree structure; Secondly, the regularization term is implanted in the objective function to minimize the structural risk. Third, a weighted segmentation criterion based on feature distribution is designed.

4. Experiment and results

4.1. Consumer purchasing behavior data selection

Since the selected e-commerce platform consumer purchase data information is large, in order to further analyze the data, this section adopts the way of database to show the selected data set.

(1) Consumer data set of an e-commerce platform (including 105,321 consumers), as shown in Table 1.

Table 1. Consumer data

User_id	User ID	Data desensitization
Age	Age group	-1 indicates unknown
Sex	sex	0 means male, 1 means female, and 2 means confidential
User_iv_cd	Consumer class	Sequential enumeration of levels. The higher the level, the larger the number
User_reg_tm	Consumer platform registration date	Grain size to day

(2) An e-commerce platform predicts consumer purchase data set (including 24,187 purchase records), as shown in Table 2.

Table 2. Data set of goods purchased by consumers

Sku_id	Product number	Data desensitization
A1	Stats 1	Enumeration, -1 indicates unknown
A2	Stats 2	Enumeration, -1 indicates unknown
A3	Stats 3	Enumeration, -1 indicates unknown
Cate	Commodity category	hyposensitization

Brand	Commodity brand	hyposensitization

(3) Review data of consumers of an e-commerce platform on products (including 558,552 review records), as shown in Table 3.

Table 3. Data of consumers' comments on products

dt	Comment deadline	Grain size to day
Sku_id	Product number	hyposensitization
Comment_num	Cumulative number of comment segments	0 means no comments, 1 means 1 comment, 2 means 2-10 comments, 3 means 11-50 comments, and 4 means more than 50 comments
Has_bad_comment	Whether there are bad reviews	0 means none, 1 means yes
Bad_comment_rate	Product negative rating rate	The proportion of negative reviews to the total number of reviews

(4) The interactive data of consumer online shopping behavior is shown in Table 4.

Table 4 Interactive data of consumer online shopping behavior

User_id	Comment deadline	hyposensitization
Sku_id	Product number	hyposensitization
Time	Interaction time	
Model_id	Click the module number, if so click	hyposensitization
Type	1. Browse (refer to browse the product details page); 2. Add to cart; 3. Delete shopping cart; 4. Place an order; 5. Pay attention; Step 6 Click	
Cate	Commodity category	hyposensitization
Brand	Commodity brand	hyposensitization

4.2. Verification of consumer online shopping data set

Based on the XGBoost model, this experiment aims to divide the existing data set of consumers' online shopping behavior into a training set and a test set by collecting relevant data of consumers through data preprocessing. On this basis, experiments are carried out through the method of cross-checking, so as to evaluate the performance of the model. A three-stage strategy is required to optimize the model performance. First, a parameter regulation system is established, with emphasis on the configuration of tree structure control parameters (such as maximum depth and leaf node weight threshold) and training process parameters (such as learning rate and feature subsampling proportion), and the basic model framework is constructed. Secondly, K-fold cross-validation is used to optimize the hyperparameter space, and the optimal combination of parameters is determined by the validation set loss function trajectory analysis. Finally, multiple rounds of iterative fitting are performed on the training set, and early stop mechanism and regularization constraint are applied simultaneously to prevent overfitting.

In the phase of model deployment, a dual safeguard mechanism should be implemented: on the one hand, the interpretability of the model should be improved through feature importance analysis, and on the other hand, a real-time monitoring system should be built to track the predicted deviation. Through the incremental update strategy of online learning and the dynamic adjustment of model parameters combined with the feedback data of business scenarios, the continuous adaptive optimization of the prediction system under the changing environment of data distribution is realized. In this process, the synergistic effect of structural parameters and regularization parameters, as well as the design of dynamic resource allocation mechanism, are the key technical supports to maintain the high predictive efficiency of the model.

The comparison of data before and after the experiment is shown in Table 5.

Table 5. Comparison of experimental results

Model class	MSE	MAE
Linear regression	6.79	3.8
Ridge regression	1	1.22
Lasso return	6	2.312

As can be seen from the above table, MSE has the largest difference in predicting consumer purchase behavior among the three model categories, while Ridge regression model has the best performance and shorter calculation time than the other two methods.

4.3. Analysis of prediction model results

In this study, a linear regression optimization framework integrating time series analysis was designed to transform the biweekly observation data into dynamic feature vectors through feature engineering to build a weekly granularity user consumption prediction model. The output result of the model contains the predicted value and confidence interval of the next seven days, which provides triple value support for the operation decision of e-commerce: the platform can build an intelligent recommendation framework driven by consumer behavior based on the prediction of consumer trends; The merchant side can not only formulate differentiated promotion strategies according to the purchasing tendency of users, implement targeted preferential incentives for groups with low conversion intention, but also dynamically adjust inventory water level combined with demand forecasting to achieve the optimization of supply and demand balance. Through model verification and final model evaluation, the accuracy rate of the model is 0.91, indicating that the model can scientifically and accurately predict consumers' purchase behavior intention on the e-commerce platform in most cases. The core innovation of the model lies in deeply coupling the traditional regression model with the business timing characteristics, and capturing the fluctuation rule of the consumption cycle through the sliding window mechanism. This forecast-driven operation model not only reduces the waste of resources in traditional wide-net marketing, but also greatly improves inventory turnover efficiency through accurate demand prediction, forming a complete value closed-loop from data insight to business decision.

5. Conclusion

The progress of society and the development of science and technology are changing the consumption behavior of contemporary people. Online consumption is the most popular consumption mode at this stage. Judging from the consumption habits, living needs and habits of contemporary people, online shopping will remain the mainstream consumption mode of society for a long time to come, and e-commerce platforms will not be used for traditional offline sales platforms. Its operating principle requires that it must accurately grasp the purchase intention of consumers, and can use regression analysis to grasp the purchase intention of consumers in the analysis of commodity distribution and after-sales problems, so as to choose better marketing strategies.

This paper develops a consumer-product matching prediction system by constructing a machine learning framework for multi-source data fusion. Based on the intelligent recommendation algorithm engine, the system establishes the dynamic correlation graph between user behavior characteristics and commodity attributes to achieve accurate demand prediction and directional recommendation. With the help of machine learning algorithm model, the e-commerce platform can integrate user portrait analysis and real-time behavior tracking to form a consumer purchase intention prediction model with temporal characteristics, so as to more efficiently and accurately predict the possible purchase intention of consumers in the future consumption cycle, and at the same time effectively identify the potential purchase demand in the future consumption cycle. On this basis, the e-commerce platform can launch intelligent replenishment strategies for high-demand forecast commodities and avoid the risk of stock shortage by presetting storage resources in advance. For low-heat goods, the inventory elastic shrinkage program is implemented to reduce unsalable losses.

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